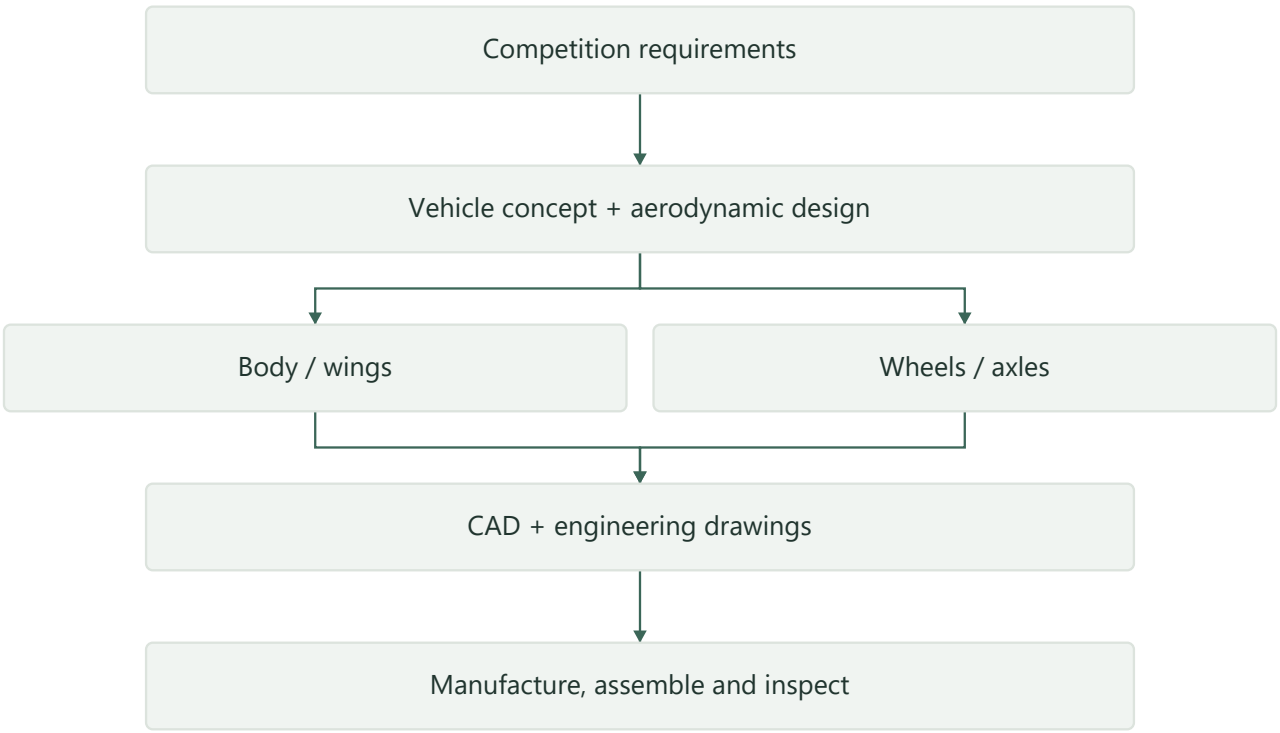
Maverick Motorsports

Miniature race-car design, aerodynamics and manufacturing

A team design-and-engineering portfolio documenting a miniature racing vehicle, with aerodynamic development, component decisions, manufacturing planning and a separate engineering-drawing set.

Architecture



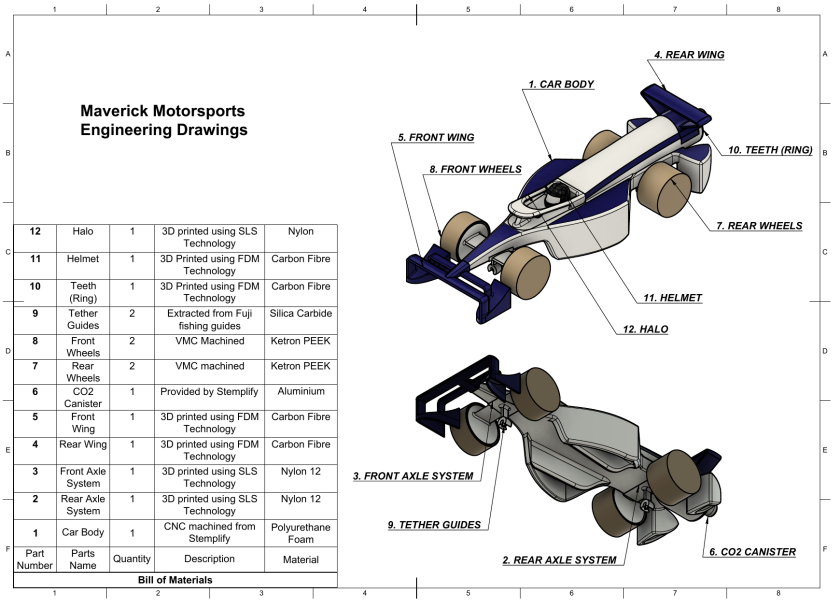
Component overview derived from the supplied source; version-specific details may differ.

Component	Implementation	Role
Vehicle design	Original design and engineering portfolio	Requirements and iteration
Drawing set	Component and assembly sheets	Dimensions and manufacture
Aerodynamics	Reported simulation comparisons	Interpretation with setup and unit checks
Manufacturing	Materials and process choices	Design-to-build reasoning

Team engineering portfolio

Vehicle layout & bill of materials

Maverick Motorsports develops a CO2-propelled STEM Racing car through iterative CAD, aerodynamic analysis, manufacturing preparation and physical scrutineering. The drawing pack provides the component-level manufacturing view.



Part Number	Parts Name	Quantity	Description	Material
12	Halo	1	3D printed using SLS Technology	Nylon
11	Helmet	1	3D Printed using FDM Technology	Carbon Fibre
10	Teeth (Ring)	1	3D Printed using FDM Technology	Carbon Fibre
9	Tether Guides	2	Extracted from Fuji fishing guides	Silica Carbide
8	Front Wheels	2	VMC Machined	Ketron PEEK
7	Rear Wheels	2	VMC machined	Ketron PEEK
6	CO2 Canister	1	Provided by Stemplify	Aluminium
5	Front Wing	1	3D printed using FDM Technology	Carbon Fibre
4	Rear Wing	1	3D printed using FDM Technology	Carbon Fibre
3	Front Axle System	1	3D printed using SLS Technology	Nylon 12
2	Rear Axle System	1	3D printed using SLS Technology	Nylon 12
1	Car Body	1	CNC machined from Stemplify	Polyurethane Foam

Bill of Materials

STEM Design PDF pp.2, 8-11; Engineering Drawings PDF p.1. Carbon-fibre process descriptions are reproduced as team terminology, not a verified laminate specification.

Assembly and materials

The BOM specifies a polyurethane-foam CNC body; SLS Nylon 12 front/rear axle systems; printed carbon-fibre wing, helmet and tooth-ring parts; VMC-machined Ketron PEEK wheels; an aluminium CO2 canister; silica-carbide tether guides and a printed Nylon halo.

Engineering objectives

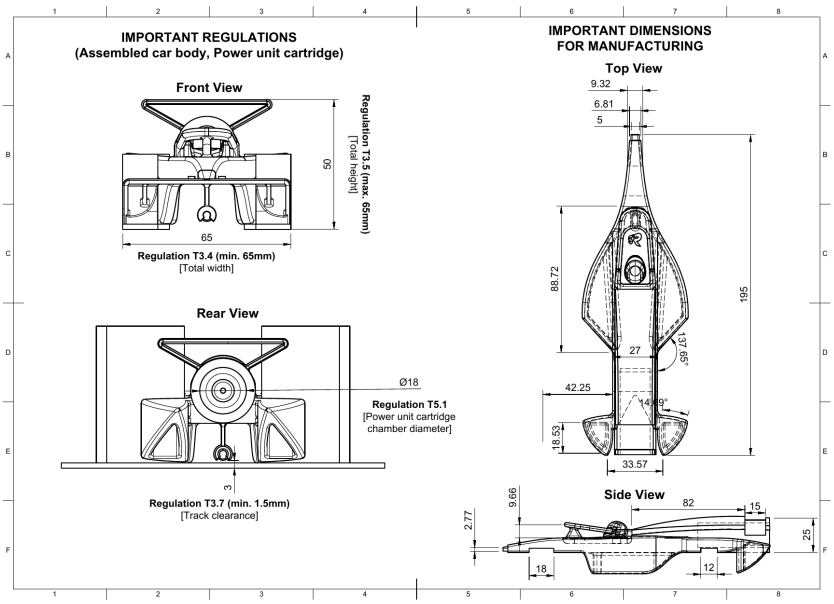
The design targets low mass near 50 g, controlled mass distribution, reduced drag and legal component envelopes. The portfolio describes a goal of improving 1.3 s to 1.1 s, while the later 1.06 s result is a simulation prediction, not a supplied track timing.

Engineering detail	Interpretation
Interfacing	Wing, axle and body attachment cavities locate outsourced parts
Provenance	Team engineering documents; individual authorship not assigned by these pages

Source / interpretation: STEM Design PDF pp.2, 8-11; Engineering Drawings PDF p.1. Carbon-fibre process descriptions are reproduced as team terminology, not a verified laminate specification.

Dimensioned layout & regulatory envelope

The drawing suite separates overall manufacture dimensions from regulation callouts and section views. These sheets should remain available at full resolution rather than be reduced to a decorative thumbnail.



Engineering Drawings PDF pp.2-5. The source calls T3.4 width min.65 mm while the manufacturing narrative discusses a 65 mm block limit; preserve rather than silently reconcile.

Assembly dimensions

The assembled drawing identifies 195 mm overall length, 65 mm width and 50 mm height. A later wing-layout sheet identifies 121 mm wheelbase, with a 120 mm minimum callout; the canister chamber is dimensioned Ø18 mm and track clearance is marked 1.5 mm minimum.

Section-view reasoning

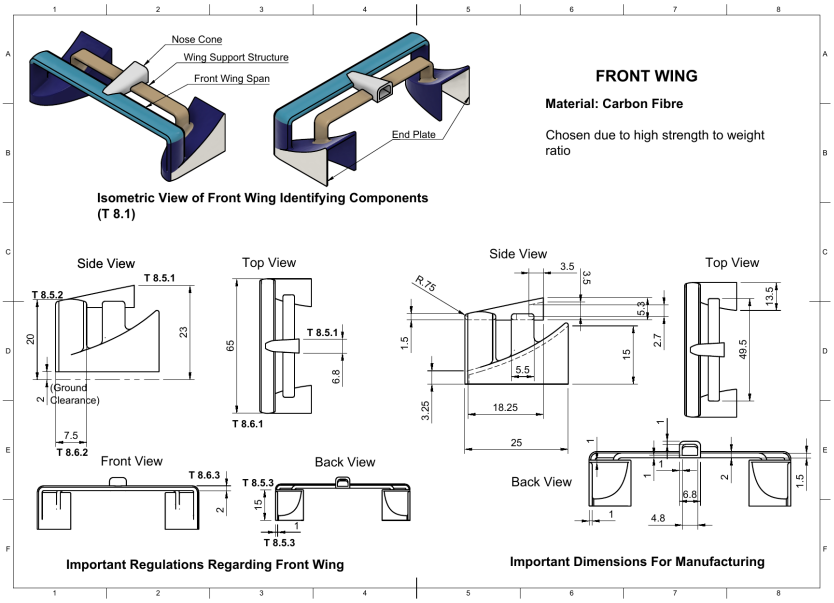
Virtual-cargo sections establish protected internal envelopes and visibility relationships. Separate sheets dimension halo visibility, canister position and protrusion, tether guide bore and wheel visibility. These are the source drawing's rule labels, not a fresh certification against current competition rules.

Engineering detail	Interpretation
Drawing use	Locate datum planes, reference lines and assembly envelopes
Caution	Width callouts and design manufacturing limit should be checked against the event rule edition

Source / interpretation: Engineering Drawings PDF pp.2-5. The source calls T3.4 width min.65 mm while the manufacturing narrative discusses a 65 mm block limit; preserve rather than silently reconcile.

Front and rear wing geometry

The wing drawings connect aerodynamic intent to actual supporting geometry, material and manufacture dimensions. The front sheet identifies nose cone, support structure, span and end plates in an isometric assembly.



Design PDF pp.4-5, 8; Engineering Drawings PDF pp.6-7. Sheet labels retain original component terminology.

Front-wing design

The development account describes reducing premature separation with smoother ramps and shelling to reduce mass, while maintaining regulation envelopes. The front drawing supplies a 65 mm span callout, 7.5 mm aerodynamic-section callout and manufacturing sections that expose wall and support geometry.

Rear-wing design

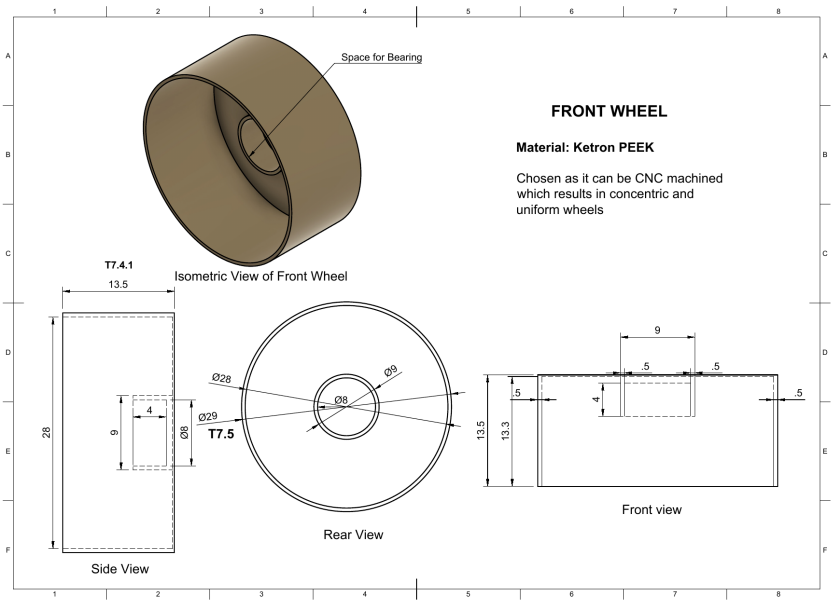
The rear design uses a swept form to reduce frontal area and wake. Its drawing gives a 50.5 mm span and 11.5 mm aerodynamic chord callout, with fillets, section profiles and thickness dimensions. Use the companion rear sheet for actual manufacture; the two wing geometries should not be merged into one generic aerofoil diagram.

Engineering detail	Interpretation
Manufacture	Printed carbon-fibre parts, described as FDM in the BOM
Verification	Dimensioned drawing plus component-specific CFD views

Source / interpretation: Design PDF pp.4-5, 8; Engineering Drawings PDF pp.6-7. Sheet labels retain original component terminology.

Wheel, bearing & axle interfaces

The wheel and axle sheets reveal how the design translates low rotational inertia and repeatable alignment into machinable interfaces. The wheels are specified in Ketron PEEK and the axle systems in Nylon 12.



Engineering Drawings PDF pp.8-11; Design PDF pp.5, 10-11. Interpret dimensions from sheets; flag radius/diameter inconsistency.

Wheel geometry

Front and rear wheel sheets show Ø29 mm outer wheel size and Ø8/Ø9 mm bearing-interface dimensions. Front and rear widths differ: the sheets label 13.5 mm and 17.5 mm respectively, with additional recess depths and fillets. Those interfaces require the full drawing, not inferred clearances from a render.

Axle manufacturing

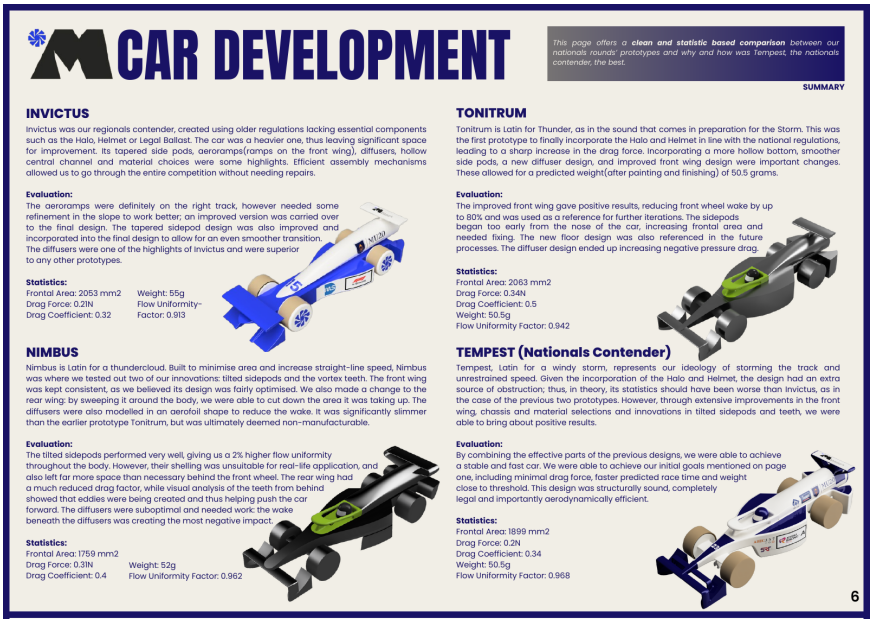
The axle systems are SLS printed and integrate wheel attachment features and tether-guide slots. The design narrative compares alternatives and selects materials for mass, flexibility, friction and manufacturing precision. The source also contains a wheel-radius typo: axle-sheet prose calls 29 mm a radius while wheel sheets dimension Ø29 mm.

Engineering detail	Interpretation
Interface control	Bearing bore, wheel recess and attachment geometry
Process control	VMC concentricity for wheels; print tolerance and fit for axle parts

Source / interpretation: Engineering Drawings PDF pp.8-11; Design PDF pp.5, 10-11. Interpret dimensions from sheets; flag radius/diameter inconsistency.

Iterations & component decisions

The design narrative records four national-development prototypes: Invictus, Nimbus, Tonitrum and Tempest. The iteration sheet explains what was retained or rejected, making the final model traceable to component-level choices.



CAR DEVELOPMENT

This page offers a **clean and statistic based comparison** between our nationals rounds' prototypes and why and how was Tempest, the nationals contender, the best.

SUMMARY

INVICTUS

Invictus was our regionals contender, created using older regulations lacking essential components such as the Halo, Helmet or Legal Ballast. The car was a heavier one, thus leaving significant space for improvement. Its tapered side pods, aeroramps(ramps on the front wing), diffusers, hollow central channel and material choices were some highlights. Efficient assembly mechanisms allowed us to go through the entire competition without needing repairs.

Evaluation:

The aeroramps were definitely on the right track, however needed some refinement in the slope to work better; an improved version was carried over to the final design. The tapered sidepod design was also improved and incorporated into the final design to allow for an even smoother transition. The diffusers were one of the highlights of Invictus and were superior to any other prototypes.

Statistics:

Frontal Area: 2053 mm2
Drag Force: 0.21N
Drag Coefficient: 0.32
Weight: 55g
Flow Uniformity-Factor: 0.913

TONITRUM

Tonitrum is Latin for Thunder, as in the sound that comes in preparation for the Storm. This was the first prototype to finally incorporate the Halo and Helmet in line with the national regulations, leading to a sharp increase in the drag force. Incorporating a more hollow bottom, smoother side pods, a new diffuser design, and improved front wing design were important changes. These allowed for a predicted weight(after painting and finishing) of 50.5 grams.

Evaluation:

The improved front wing gave positive results, reducing front wheel wake by up to 80% and was used as a reference for further iterations. The sidepods began too early from the nose of the car, increasing frontal area and needed fixing. The new floor design was also referenced in the future processes. The diffuser design ended up increasing negative pressure drag.

Statistics:

Frontal Area: 2063 mm2
Drag Force: 0.34N
Drag Coefficient: 0.5
Weight: 50.5g
Flow Uniformity Factor: 0.942

NIMBUS

Nimbus is Latin for a thundercloud. Built to minimise area and increase straight-line speed, Nimbus was where we tested out two of our innovations: tilted sidepods and the vortex teeth. The front wing was kept consistent, as we believed its design was fairly optimised. We also made a change to the rear wing: by sweeping it around the body, we were able to cut down the area it was taking up. The diffusers were also modelled in an aerofoil shape to reduce the wake. It was significantly slimmer than the earlier prototype Tonitrum, but was ultimately deemed non-manufacturable.

Evaluation:

The tilted sidepods performed very well, giving us a 2% higher flow uniformity throughout the body. However, their shelling was unsuitable for real-life application, and also left far more space than necessary behind the front wheel. The rear wing had a much reduced drag factor, while visual analysis of the teeth from behind showed that eddies were being created and thus helping push the car forward. The diffusers were suboptimal and needed work: the wake beneath the diffusers was creating the most negative impact.

Statistics:

Frontal Area: 1759 mm2
Drag Force: 0.31N
Drag Coefficient: 0.4
Weight: 52g
Flow Uniformity Factor: 0.962

TEMPEST (Nationals Contender)

Tempest, Latin for a windy storm, represents our ideology of storming the track and unrestrained speed. Given the incorporation of the Halo and Helmet, the design had an extra source of obstruction; thus, in theory, its statistics should have been worse than Invictus, as in the case of the previous two prototypes. However, through extensive improvements in the front wing, chassis and material selections and innovations in tilted sidepods and teeth, we were able to bring about positive results.

Evaluation:

By combining the effective parts of the previous designs, we were able to achieve a stable and fast car. We were able to achieve our initial goals mentioned on page one, including minimal drag force, faster predicted race time and weight close to threshold. This design was structurally sound, completely legal and importantly aerodynamically efficient.

Statistics:

Frontal Area: 1899 mm2
Drag Force: 0.2N
Drag Coefficient: 0.34
Weight: 50.5g
Flow Uniformity Factor: 0.968

6

Design & Engineering PDF pp.3-8. Results are portfolio-reported simulation outputs and design evaluations.

Tradeoffs

Sidepod tilt and hollowing were explored to control flow and mass. Tonitrum is explicitly described as non-manufacturable despite promising digital behavior. Tempest combines selected earlier features with revised diffuser and rear geometry, while addition of the halo/helmet increased frontal area.

Interpreting comparison figures

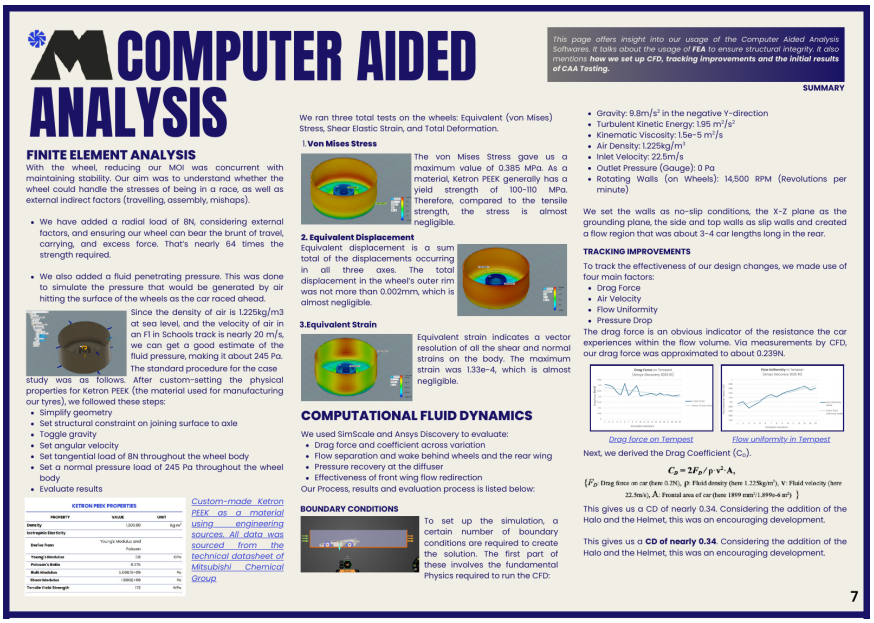
The final table reports Tempest frontal area 1899 mm^2, drag coefficient 0.34, mass 50.5 g and flow-uniformity factor 0.968. The drag force is listed as 0.2 N on this sheet and approximately 0.239 N in the analysis chapter; preserve these as source-reported values with setup/version context, not one exact reproduced result.

Engineering detail	Interpretation
Design learning	A numerically attractive concept can fail manufacturing constraints
Evidence	Versioned CAD and matched simulation setup are needed for fair comparison

Source / interpretation: Design & Engineering PDF pp.3-8. Results are portfolio-reported simulation outputs and design evaluations.

CFD and structural analysis setup

The analysis chapter documents both aerodynamic and wheel-structure simulations. Their boundary conditions are more informative than an isolated drag or stress number, and should accompany any displayed result.



COMPUTER AIDED ANALYSIS

This page offers insight into our usage of the Computer Aided Analysis Softwares. It talks about the usage of **FEA** to ensure structural integrity. It also mentions **how we set up CFD, tracking improvements and the initial results of CAA Testing.**

SUMMARY

We ran three total tests on the wheels: Equivalent (von Mises) Stress, Shear Elastic Strain, and Total Deformation.

FINITE ELEMENT ANALYSIS

With the wheel, reducing our MOI was concurrent with maintaining stability. Our aim was to understand whether the wheel could handle the stresses of being in a race, as well as external indirect factors (travelling, assembly, mishaps).

- We have added a radial load of 8N, considering external factors, and ensuring our wheel can bear the brunt of travel, carrying, and excess force. That's nearly 64 times the strength required.
- We also added a fluid penetrating pressure. This was done to simulate the pressure that would be generated by air hitting the surface of the wheels as the car raced ahead.

Since the density of air is 1.225kg/m3 at sea level, and the velocity of air in an F1 in Schools track is nearly 20 m/s, we can get a good estimate of the fluid pressure, making it about 245 Pa. The standard procedure for the case study was as follows. After custom-setting the physical properties for Ketron PEEK (the material used for manufacturing our tyres), we followed these steps:

- Simplify geometry
- Set structural constraint on joining surface to axle
- Toggle gravity
- Set angular velocity
- Set tangential load of 8N throughout the wheel body
- Set a normal pressure load of 245 Pa throughout the wheel body
- Evaluate results

KETRON PEEK PROPERTIES

PROPERTY	VALUE	UNIT
Density	1,320.00	kg/m³
Isotropic Elasticity		
Define From	"Young's Modulus and Poisson"	
Young's Modulus	3.81	GPa
Poisson's Ratio	0.375	
Bulk Modulus	3.9583E+09	Pa
Shear Modulus	1.3900E+09	Pa
Tensile Yield Strength	172	MPa

Custom-made Ketron PEEK as a material using engineering sources. All data was sourced from the technical datasheet of Mitsubishi Chemical Group

1. Von Mises Stress

The von Mises Stress gave us a maximum value of 0.385 MPa. As a material, Ketron PEEK generally has a yield strength of 100-110 MPa. Therefore, compared to the tensile strength, the stress is almost negligible.

2. Equivalent Displacement

Equivalent displacement is a sum total of the displacements occurring in all three axes. The total displacement in the wheel's outer rim was not more than 0.002mm, which is almost negligible.

3.Equivalent Strain

Equivalent strain indicates a vector resolution of all the shear and normal strains on the body. The maximum strain was 1.33e-4, which is almost negligible.

COMPUTATIONAL FLUID DYNAMICS

We used SimScale and Ansys Discovery to evaluate:

- Drag force and coefficient across variation
- Flow separation and wake behind wheels and the rear wing
- Pressure recovery at the diffuser
- Effectiveness of front wing flow redirection

Our Process, results and evaluation process is listed below:

BOUNDARY CONDITIONS

To set up the simulation, a certain number of boundary conditions are required to create the solution. The first part of these involves the fundamental Physics required to run the CFD:

- Gravity: 9.8m/s² in the negative Y-direction
- Turbulent Kinetic Energy: 1.95 m²/s²
- Kinematic Viscosity: 1.5e-5 m²/s
- Air Density: 1.225kg/m³
- Inlet Velocity: 22.5m/s
- Outlet Pressure (Gauge): 0 Pa
- Rotating Walls (on Wheels): 14,500 RPM (Revolutions per minute)

We set the walls as no-slip conditions, the X-Z plane as the grounding plane, the side and top walls as slip walls and created a flow region that was about 3-4 car lengths long in the rear.

TRACKING IMPROVEMENTS

To track the effectiveness of our design changes, we made use of four main factors:

- Drag Force
- Air Velocity
- Flow Uniformity
- Pressure Drop

The drag force is an obvious indicator of the resistance the car experiences within the flow volume. Via measurements by CFD, our drag force was approximated to about 0.239N.

Drag force on Tempest *Flow uniformity in Tempest*

Next, we derived the Drag Coefficient (C_D).

$$C_D = 2F_D / \rho \cdot v^2 \cdot A,$$

$\{F_D$: Drag force on car (here 0.2N), ρ: Fluid density (here 1.225kg/m³), v: Fluid velocity (here 22.5m/s), A: Frontal area of car (here 1899 mm²/1.899e-6 m²) $\}$

This gives us a CD of nearly 0.34. Considering the addition of the Halo and the Helmet, this was an encouraging development.

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7

Design PDF pp.7-8. Source statements are retained as reported simulations; no independent solver rerun was conducted.

Aerodynamic model

The portfolio uses SimScale and Ansys Discovery, with 22.5 m/s inlet speed, density 1.225 kg/m^3, kinematic viscosity 1.5e-5 m^2/s, zero-gauge outlet pressure, wheel rotation 14,500 RPM, no-slip car walls and slip outer boundaries. It later identifies k-omega SST and reports y+ about 0.5.

Wheel loading

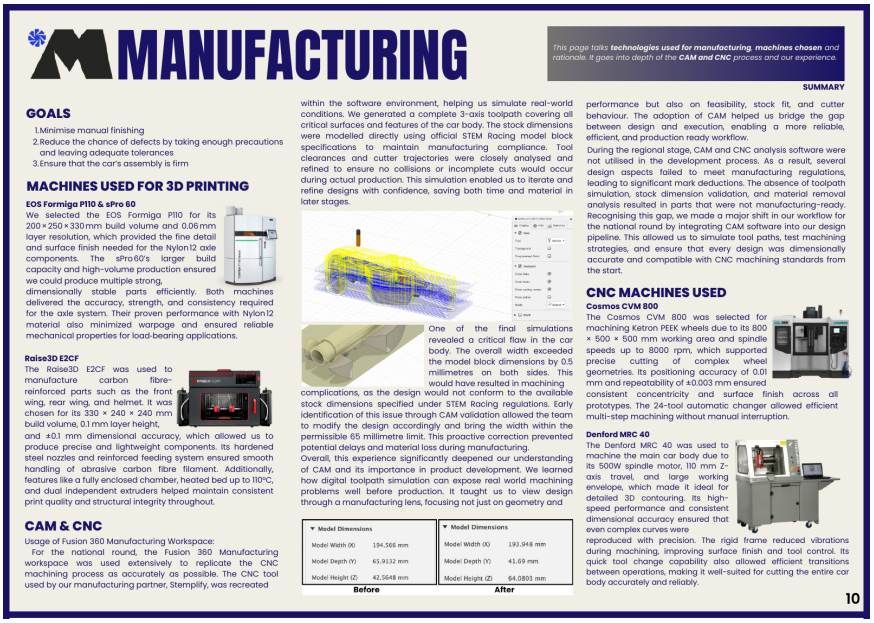
The wheel study constrains the axle joining surface, includes gravity/angular velocity, applies an 8 N load and 245 Pa pressure, and reports 0.385 MPa maximum stress, ≤0.002 mm displacement and 1.33e-4 strain. The narrative alternates radial and tangential language for 8 N, so its exact load direction needs the solver file.

Engineering detail	Interpretation
CFD qualification	A low y+ alone does not demonstrate mesh independence or solver accuracy
FEA qualification	Small reported stress depends on constraints, material data and load direction

Source / interpretation: Design PDF pp.7-8. Source statements are retained as reported simulations; no independent solver rerun was conducted.

CAD to CAM and manufacturing reform

Manufacturing feedback changed the design process between regionals and nationals. Instead of discovering fit problems during sanding, the team incorporated stock geometry, toolpath simulation and process clearances into the CAD workflow.



MANUFACTURING

This page talks **technologies used for manufacturing**, **machines chosen** and rationale. It goes into depth of the **CAM and CNC** process and our experience.

SUMMARY

GOALS

1. Minimise manual finishing
2. Reduce the chance of defects by taking enough precautions and leaving adequate tolerances
3. Ensure that the car's assembly is firm

MACHINES USED FOR 3D PRINTING

EOS Formiga P110 & sPro 60

We selected the EOS Formiga P110 for its 200 × 250 × 330 mm build volume and 0.06 mm layer resolution, which provided the fine detail and surface finish needed for the Nylon12 axle components. The sPro60's larger build capacity and high-volume production ensured we could produce multiple strong, dimensionally stable parts efficiently. Both machines delivered the accuracy, strength, and consistency required for the axle system. Their proven performance with Nylon12 material also minimized warpage and ensured reliable mechanical properties for load-bearing applications.

Raise3D E2CF

The Raise3D E2CF was used to manufacture carbon fibre-reinforced parts such as the front wing, rear wing, and helmet. It was chosen for its 330 × 240 × 240 mm build volume, 0.1 mm layer height, and ±0.1 mm dimensional accuracy, which allowed us to produce precise and lightweight components. Its hardened steel nozzles and reinforced feeding system ensured smooth handling of abrasive carbon fibre filament. Additionally, features like a fully enclosed chamber, heated bed up to 110°C, and dual independent extruders helped maintain consistent print quality and structural integrity throughout.

CAM & CNC

Usage of Fusion 360 Manufacturing Workspace:

For the national round, the Fusion 360 Manufacturing workspace was used extensively to replicate the CNC machining process as accurately as possible. The CNC tool used by our manufacturing partner, Stemplify, was recreated within the software environment, helping us simulate real-world conditions. We generated a complete 3-axis toolpath covering all critical surfaces and features of the car body. The stock dimensions were modelled directly using official STEM Racing model block specifications to maintain manufacturing compliance. Tool clearances and cutter trajectories were closely analysed and refined to ensure no collisions or incomplete cuts would occur during actual production. This simulation enabled us to iterate and refine designs with confidence, saving both time and material in later stages.

One of the final simulations revealed a critical flaw in the car body. The overall width exceeded the model block dimensions by 0.5 millimetres on both sides. This would have resulted in machining complications, as the design would not conform to the available stock dimensions specified under STEM Racing regulations. Early identification of this issue through CAM validation allowed the team to modify the design accordingly and bring the width within the permissible 65 millimetre limit. This proactive correction prevented potential delays and material loss during manufacturing.

Overall, this experience significantly deepened our understanding of CAM and its importance in product development. We learned how digital toolpath simulation can expose real world machining problems well before production. It taught us to view design through a manufacturing lens, focusing not just on geometry and performance but also on feasibility, stock fit, and cutter behaviour. The adoption of CAM helped us bridge the gap between design and execution, enabling a more reliable, efficient, and production ready workflow.

During the regional stage, CAM and CNC analysis software were not utilised in the development process. As a result, several design aspects failed to meet manufacturing regulations, leading to significant mark deductions. The absence of toolpath simulation, stock dimension validation, and material removal analysis resulted in parts that were not manufacturing-ready. Recognising this gap, we made a major shift in our workflow for the national round by integrating CAM software into our design pipeline. This allowed us to simulate tool paths, test machining strategies, and ensure that every design was dimensionally accurate and compatible with CNC machining standards from the start.

Model Dimensions		Model Dimensions	
Model Width (X)	194.566 mm	Model Width (X)	193.948 mm
Model Depth (Y)	65.9132 mm	Model Depth (Y)	41.69 mm
Model Height (Z)	42.5648 mm	Model Height (Z)	64.0803 mm
Before		After	

CNC MACHINES USED

Cosmos CVM 800

The Cosmos CVM 800 was selected for machining Ketron PEEK wheels due to its 800 × 500 × 500 mm working area and spindle speeds up to 8000 rpm, which supported precise cutting of complex wheel geometries. Its positioning accuracy of 0.01 mm and repeatability of ±0.003 mm ensured consistent concentricity and surface finish across all prototypes. The 24-tool automatic changer allowed efficient multi-step machining without manual interruption.

Denford MRC 40

The Denford MRC 40 was used to machine the main car body due to its 500W spindle motor, 110 mm Z-axis travel, and large working envelope, which made it ideal for detailed 3D contouring. Its high-speed performance and consistent dimensional accuracy ensured that even complex curves were reproduced with precision. The rigid frame reduced vibrations during machining, improving surface finish and tool control. Its quick tool change capability also allowed efficient transitions between operations, making it well-suited for cutting the entire car body accurately and reliably.

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Design PDF pp.9-11. Machine specifications are source claims; no current vendor specification verification is implied.

Digital manufacture

Fusion 360 Manufacturing models the Stemplify CNC process, including a 3-axis path and official block geometry. A simulation exposed body width extending 0.5 mm beyond each side of the stock. Correcting the width before machining avoided material loss and rework.

Process choices

The portfolio assigns body machining to Denford MRC 40, PEEK wheel machining to Cosmos CVM 800, and Nylon 12 SLS work to EOS Formiga P110/sPro 60. Raise3D E2CF is listed for carbon-fibre-reinforced printed parts. Tool access uses fillets and a 2.5 mm end mill; print design uses 45-degree overhang limits and stated 0.8 mm SLS/1.5 mm FDM walls.

Engineering detail	Interpretation
Design rule	Fit within stock and allow cutter access before final geometry
Supplier interface	Stemplify, Makenica, Micro Tools and Services, Shree Rapid Technologies

Source / interpretation: Design PDF pp.9-11. Machine specifications are source claims; no current vendor specification verification is implied.

Assembly, validation & predicted race time

The final assembly workflow joins outsourced precision parts to a finished foam body while retaining dimension-critical attachment surfaces. The validation record combines physical scrutineering with simulation rather than a supplied regulation-track test.



MANUFACTURING

*This page talks about the **Assembly and Final finishing process** including surface finishing, painting, glueing etc. It also mentions **Workplace Safety** measures and **Quality Analysis:** How we ensured our car model met our standards.*

SUMMARY

MANUFACTURING REFORMS

At the regional stage, no tolerances were applied for CNC machining or 3D printing, resulting in poor fits and extensive manual sanding during assembly. This increased post-processing time and introduced risks of dimensional inaccuracies. For the nationals, we referred to the specific tolerance limits of each machine model and incorporated clearances accordingly. This ensured better part compatibility, reduced manual work, and improved overall manufacturing accuracy. The regional car body design included extremely thin fillets and narrow joints, making it structurally weak and prone to damage during CNC machining. In some cases, this could have caused the model block to crack or break. For the updated design, we increased joint thickness and used smoother transitions to strengthen the structure and ensure safer, more reliable manufacturing.

ASSEMBLY & WORKPLACE SAFETY

Outsourced materials:
The axle system and hub caps, made of Nylon 12, were outsourced to Makenica since our main manufacturer, Stemplify, did not support this material. The outsourcing also helped reduce costs due to Makenica's competitive pricing.
The front and rear wings, made of carbon fibre, were provided free of cost by our aid sponsor, Shree Rapid Technologies, who had access to the correct grade suitable for our use.
The wheels, made from Ketron PEEK, were outsourced to Micro Tools and Services. As PEEK is a specialised, imported plastic, it requires highly precise VMC machining with very low tolerances, which the vendor in India successfully carried out.

Sanding & Finishing:
After CNC machining, the car body was sanded to ensure a smooth and even finish. Starting with 320-grit sandpaper to remove surface defects, we progressed to 2000-grit for final polishing. An electric hand sander was used for precision and consistency, especially in detailed areas.
Carbon fibre and Nylon 12 parts, made via 3D printing, also had surface irregularities and uneven edges, which were manually sanded for improved fit and finish. Although the VMC-machined wheels were highly precise, minor edge inconsistencies were corrected using 1000-grit sandpaper to ensure smooth rotation and balance.

Painting & Sealing:
- Each part of the car was painted individually to reduce weight by avoiding hidden areas and to ensure greater precision. The entire primer and paint process was carried out by a professional company, Lords of Detailing, equipped with specialised tools for a smooth and uniform finish.
- Before priming, the car body was sealed with a mix of Fevicol and water to prevent polyurethane foam from reacting with hydrocarbon-based primers and paints. A coat of plastic primer was then applied to all parts and left to dry for 2 hours, chosen for its strong adhesion and protection against corrosion and cracking.
- Spray paint was applied according to the colour scheme, using masking tape to separate sections. After drying, waterslide decals formed the livery, and vinyl stickers were used for the 'STEM Racing' logo, as per regulations. A clear coat was added to seal the surface and protect it from moisture. No paint or primer was applied to attachment spaces, as they were dimension-critical.

Assembly:
After painting, Loctite SF 770 primer was applied to all joints and left for 1 minute to activate the inert Nylon 12 surfaces. Loctite 406 was used for larger joints, while Loctite Precision was applied to small gaps like wheels and bearings. This ensured strong, reliable bonding. Attachment spaces were included in the car body, with space left for glue to securely fix outsourced parts like the front wing, rear wing, and wheel system.

WORKPLACE SAFETY:

Risk	Risk Cause	Risk Control
Skin irritation, Breathing problems	Spilling of super glue, inhalation of spray paint	Medical masks and high-grade gloves were used
Hand injuries and cuts	Accidental contact with sanding surfaces or sharp parts	Use of clamps and blocks to hold components securely while sanding
Eye injury	Flying dust or particles during sanding or drilling	Use of protective safety goggles at all times during finishing processes

Manufacturing & Assembly

Physical Testing

QUALITY ANALYSIS

Without access to a regulation F1 in Schools track, our team had to approach physical testing through alternative yet effective methods. Our process was centred around three major components.
First, we rigorously monitored and **adjusted the car's weight** throughout development to ensure it consistently met the target, a crucial factor for maximising acceleration and maintaining structural efficiency. Considering we needed only minor fine-tuning, we opted **not to use the legal ballast and relied on controlled sanding and painting** to adjust weight.
Second, we conducted in-depth **physical scrutineering using precision tools**(vernier callipers, gram weight scale, screw gauge) to confirm that all critical dimensions, angles, and tolerances matched our CAD model, reducing the risk of disqualification and ensuring aerodynamic accuracy.
Finally, to validate our digital simulations, we collaborated with SGSITS, our local engineering college, to carry out a **wind tunnel test.** This allowed us to compare our CFD predictions with real-world airflow behaviour, giving us valuable insight into the performance of our aerodynamic features and enabling final refinements to the car design.

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Design PDF pp.8, 11. No raw wind-tunnel dataset, thrust curve or measured 1.06 s track timing is supplied.

Finish and bonding

The team describes sanding from 320 to 2000 grit, 1000-grit wheel-edge correction, foam sealing before paint, and avoiding paint on attachment spaces. Loctite SF 770 primer activates Nylon 12 joints before Loctite 406 or Precision adhesive. Calipers, mass scale and screw gauge check final geometry.

Testing boundaries

The portfolio states that a regulation track was unavailable and describes wind-tunnel collaboration with SGSITS. The race-time model uses one-dimensional Euler integration, 0.0001 s steps, 20 m distance, peak thrust 10 N and a thrust decline after about 0.7 s, yielding a predicted 1.06 s. Its text incorrectly gives the drag coefficient a force unit; 0.34 is dimensionless.

Engineering detail	Interpretation
Physical evidence	Scrutineering and reported wind-tunnel comparison
Prediction	Thrust-time assumptions and solver timestep govern simulated race time

Source / interpretation: Design PDF pp.8, 11. No raw wind-tunnel dataset, thrust curve or measured 1.06 s track timing is supplied.

Evidence & source

Implementation, supporting records, and the boundary of the claim

Area	Evidence	Reference
Design record	11-page design and engineering portfolio	Supplied original PDF
Drawing record	12-page engineering drawings	Supplied original PDF
Attribution	Maverick Motorsports team	Original documents retain credits

Primary references

File / record	Path / reference
Original design and engineering portfolio	documents/stem-design-original.pdf
Original engineering drawings	documents/stem-drawings-original.pdf

Original team engineering portfolio and engineering drawings are linked in full, alongside a detailed technical companion and page-specific visual gallery.

Evidence boundary

Designs, simulations and specifications are attributed to the team documents. Dimensional and CFD inconsistencies are discussed in context; reported calculations are not silently relabeled as measured track performance.

Technical note

The companion preserves the original drawings and discusses component function, dimensions, materials, manufacturing and the evidence behind design decisions.